

KOCHKI, Russian Federation

WMO: 297240

Lat: 54.336N

Lon: 80.472E

Elev: 161

StdP: 99.41

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.6	(c) -33.2	(d) -39.6	(e) 0.1	(f) -36.4	(g) -36.2	(h) 0.1	(i) -33.0	(j) 12.5	(k) -7.0	(l) 11.2	(m) -8.5	(n) 1.1	(o) 240	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.7	18.6	28.0	18.1	26.4	17.5	20.4	26.4	19.5	25.4	18.7	24.3	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	22.9	17.4	12.7	22.0	16.5	12.0	21.5	59.6	26.2	56.6	25.4	53.9	24.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.0	7.8	-38.2	32.5	4.1	2.0	-41.2	34.0	-43.6	35.2	-45.9	36.3	-48.9	37.8
			-38.2	21.9	3.9	1.2	-41.0	22.8	-43.3	23.4	-45.4	24.1	-48.3	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-18.4	-15.9	-7.1	3.8	11.8	17.3	19.0	16.6	10.3	3.2	-7.0	-14.5
	DBStd	14.48	8.57	7.53	6.23	5.72	5.34	4.07	3.18	3.70	4.61	5.11	7.82	8.08
	HDD10.0	3922	879	726	529	200	45	3	0	1	52	217	511	759
	HDD18.3	6195	1138	959	788	437	211	66	30	77	243	469	761	1017
	CDD10.0	888	0	0	0	13	100	221	279	207	62	6	0	0
	CDD18.3	120	0	0	0	0	7	34	51	24	3	0	0	0
	CDH23.3	1258	0	0	0	5	141	347	432	277	56	0	0	0
	CDH26.7	304	0	0	0	1	34	82	103	70	15	0	0	0
Wind	WSAvg	3.2	3.4	3.4	3.5	3.4	3.3	2.8	2.4	2.5	2.8	3.2	3.7	3.9
Precipitation	PrecAvg	386	17	14	17	21	32	48	68	46	35	35	30	25
	PrecMax	531	34	24	38	48	73	84	156	100	64	81	61	48
	PrecMin	274	2	1	3	3	2	10	17	13	6	12	18	9
	PrecStd	70	9	6	9	11	18	17	36	22	17	18	11	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	0.8	6.8	23.4	30.4	31.0	31.9	31.4	28.9	19.9	8.6	1.2
		MCWB	0.0	-0.3	3.3	12.4	16.5	19.1	20.4	19.3	16.1	11.5	6.1	0.2
	2%	DB	-1.9	-1.0	4.1	19.3	26.9	29.0	29.3	28.7	24.2	16.4	5.8	0.0
		MCWB	-2.6	-1.9	1.7	10.9	14.9	18.0	19.8	18.9	14.6	9.7	4.0	-0.6
	5%	DB	-3.9	-2.8	2.7	16.2	24.5	27.3	27.5	26.5	21.4	13.3	3.7	-1.8
		MCWB	-4.7	-3.6	0.8	9.5	13.9	17.6	19.1	17.8	13.5	8.3	2.4	-2.5
	10%	DB	-5.9	-4.7	1.3	13.0	21.6	25.3	25.7	24.3	18.6	10.5	1.8	-3.7
		MCWB	-6.6	-5.5	-0.1	7.6	12.7	17.0	18.5	16.7	12.0	6.8	0.7	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	-0.3	3.6	13.5	17.8	20.4	22.1	21.2	17.3	12.1	6.5	0.4
		MCDB	0.9	0.4	6.1	21.2	27.9	27.0	28.4	27.7	24.6	18.1	8.3	0.8
	2%	WB	-2.5	-1.8	2.0	11.7	16.1	19.3	20.9	19.7	15.3	10.3	4.1	-0.6
		MCDB	-1.8	-1.1	3.8	18.2	24.8	26.2	26.6	26.2	22.5	15.1	5.6	-0.1
	5%	WB	-4.5	-3.5	1.0	9.9	14.6	18.5	20.0	18.7	13.9	8.7	2.3	-2.4
		MCDB	-3.9	-2.8	2.4	15.3	22.4	25.0	25.5	24.6	20.6	12.6	3.5	-1.8
	10%	WB	-6.6	-5.5	0.0	8.1	13.4	17.7	19.3	17.6	12.5	7.1	0.7	-4.4
		MCDB	-6.0	-4.7	1.3	12.8	20.7	23.7	24.4	23.2	17.7	10.4	1.8	-3.7
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.1	9.3	9.8	12.0	10.7	10.2	11.3	10.8	8.4	6.6	7.7
		MCDBR	8.4	8.6	8.7	15.1	16.2	14.0	12.9	14.7	16.0	13.8	6.4	7.6
		MCWBR	8.4	8.2	6.9	8.4	7.1	5.7	5.1	6.2	7.5	8.1	5.2	7.5
	5% WB	MCDBR	8.3	8.3	7.8	14.0	14.0	11.8	10.8	12.2	14.2	12.4	6.2	7.4
		MCWBR	8.3	8.0	6.4	8.3	6.9	5.5	5.0	5.8	7.1	7.9	5.3	7.4
Clear-Sky Solar Irradiance	taub		0.292	0.324	0.377	0.405	0.407	0.408	0.434	0.414	0.368	0.351	0.327	0.295
	taud		2.021	1.974	1.911	2.117	2.202	2.280	2.202	2.235	2.315	2.235	2.068	1.971
	Ebn at Noon		611	724	768	814	837	841	807	799	792	702	558	502
	Edh at Noon		73	108	146	137	133	125	133	121	98	84	70	62
All-Sky Solar Radiation	RadAvg		1.02	2.07	3.54	4.73	5.57	6.09	5.48	4.68	3.24	1.66	0.80	0.67
	RadStd		0.08	0.12	0.25	0.30	0.58	0.52	0.35	0.32	0.35	0.22	0.13	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page